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Quantum Theory of Anharmonic Oscillators

- A Variational and Systematic General Approximation Method -

Kazuo Yamazaki

Institut für Theoretische Physik,
der Universität Graz, Austria

and

Physics Deparment; College of General Education,
Kyoto University, Kyoto 606, Japan+)

+) Permanent address, and after March 1983

Abstract:

This is an investigation of the energy levels and wave functions of an anharmonic oscillator characterized by the potential $\frac{\omega^2}{2}q^2+\lambda q^4$. As a lowest order approximation an extremely simple formula for energy levels: $E_i^{(0)} = (i+\frac{1}{2})\frac{1}{4}(\frac{3}{\alpha_i}+\alpha_i)$ is derived (i being the quantum number of the energy level), which covers any (λ,i). α_i is the real positive root of a cubic equation: $y_i\alpha_i^3+\alpha_i^2-1 = 0$, with $y_i = 6\lambda(2i^2+2i+1)/(2i+1)$. This formula reproduces the exact energy levels within an error of about 1% for any (λ,i) (The worst case is 2% for $i=0$, $\lambda\to\infty$). Systematically higher orders of "our" perturbation theory are developed, which contains the "usual" perturbation theory for the limiting case of small λ, but "our" perturbation theory is valid for any (i,λ). "Our" second order perturbation theory reduces the errors of our lowest order results by a factor of about 1/5 in general. Various ranges(large, intermediate, small) of (i,λ) are investigated and compared with the exact values obtained by the Montroll group[1),2)]. For the case of $i=0,1$, even "our" forth order perturbation calculation can be elaborated explicitly, which reduces the error to about 0.01% for any λ. For small λ it gives correct numerical coefficients up to λ^4 terms, as it should be.

§1. Introduction and Summary

The anharmonic oscillator model has played a tremendous role, because it is a most simple but non trivial nonlinear problem. It occurs in the evolution of many branches of physics ranging from particle physics to molecular dynamics. For the relevance of this model to the various branches of physics we would like to refer to the paper of Hioe et al.[1]. For this reason the Montroll group elaborated the detailed papers[1),2)], using a big computer facility. We will use these papers as a standard reference for comparison with our work, and will refer to their results as e.g. (1-1.18.b), meaning reference 1 eq. (1.18b) etc.

Our way of approach to the problem consists of some kind of variational approximation, the "variational" parameter α_i is adjusted for each value of i and λ with a special combination $y_i = 6\lambda(2i^2+2i+1)/(2i+1)$. α_i is a root of eq. (2.10). y_i is essentially 6Λ in the notation of ref.1. In order to be applicable to more realistic problems, the method should be as simple as possible. Our method is simple enough, amounting to performing an appropriate scale transformation (or Bogoliubov transformation) corresponding to each given (λ,i). We decompose the canonical variables q and p into creation and annihilation operators by $q = \sqrt{\frac{\alpha_i}{2\omega}}\,(b_i+b_i^+)$, $p = \frac{1}{i}\sqrt{\frac{\omega}{2\alpha_i}}\,(b_i-b_i^+)$. $\alpha_i = 1$ corresponds to the "usual" treatment. Expressing our total Hamiltonian through b_i and b_i^+, then after the performance of Wick ordering, the diagonal term with respect to $n_i = b_i^+ b_i$ is defined as $H_i^{(0)}$ and the remaining non diagonal term as $H_i^{(I)}$. The single "variational" parameter α_i is determined such as

to make $E_i^{(0)}$ obtained from $H_i^{(0)}$ an optimum, which will make $H_i^{(I)}$ small. $E_i^{(0)}$, obtained in this way, is given by the formula in the abstract, which covers fairly well <u>all ranges</u> of (i,λ). If we plot a graph of the energy levels using this formula, we get almost the same graph as given by (1 - Fig.1). We cannot see any difference at this scale of the graphs. In order to see more clearly the accuracy of our results, we draw a graph of $R_i^{(0)} = E_i^{(0)}/E_i^{(e)}$ in Fig.4, where i is the level number, (0) means our zeroth approximation results, and (e) means the exact values of reference 1. In Fig.4 we also draw $R_i^{(2)} = (E_i^{(0)}+E_i^{(2)})/E_i^{(e)}$, where (2) means "our" 2nd order approximation results, which improves $R_i^{(0)}$ considerably.

Our $H_i^{(0)}$ is diagonal but nonlinear (containing n_i^2), and through our choice of α_i we extract out of it in a subtle way optimally nonlinear effects corresponding to each (i,λ). Our splitting into $H_i^{(0)}$ and $H_i^{(I)}$ is so differently and nicely done for each (i,λ), that although our $H_i^{(I)}$ contains no small fixed parameter, its effects are always small for any range of (i,λ). Incidentally, through our choice of α_i, our $H_i^{(I)}$ have vanishing matrix elements $\langle 2|H_0^{(I)}|0\rangle$ and $\langle 3|H_1^{(I)}|1\rangle$, for any λ as will be shown in (5.2) and (5.4), which makes "our" perturbation calculation especially simple for i=0 and i=1.

In §2 we will explain our method of approximation sketched above, and §3 contains the results for "our" lowest order approximation. From §4 on, we will develop in a systematic way "our" higher order perturbation theory, treating $H_i^{(I)}$ as perturbation, which can be done, in principle, to any order. As we have emphasized above, $H_i^{(I)}$ does <u>not</u> contain any small fixed numerical coefficient, still effectively it

is always small, as will be seen below. §4 contains "our" 2nd order perturbation results $E_i^{(2)}$,*) which reduces to the "usual" second order perturbation theory in the special limiting case of small λ, while our formula (4.4) covers all ranges of (i,λ), including very large λ. It is pretty accurate also for the intermediate values of i or λ, as will be most clearly seen in Fig.4 or Table 4.

The cases i=0 and i=1 are especially simple, so we can even go to the fourth order of "our" perturbation calculation. We do that in §5 and it reduces the error to within 0.01% for any λ, so that in Fig. 4 $R_0^{(4)}$ and $R_1^{(4)}$ lie almost on the abscissa on this scale of graphs. For small λ, as shown in (5.16), they contain the well known usual perturbation coefficients up to the λ^4 term correctly. But our general formula (5.12) and (5.17) are valid for any λ, including $\lambda\to\infty$.

In §6 we discuss the small λ regime, based on "our" 2nd order perturbation results (4.4) and arrive at (6.4), and, for large i with $(i\ \lambda) = \beta$ finite, at (6.5) which is equally good for small $\beta = \lambda i$ as the result of (1-1.18b). Their (1-1.18b) is not good for $\beta \gtrsim 0.5$, while our (4.4) is good also for any β, as will be clearly seen in Fig.5.

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*) We use the notational convention $E_i^{(j)}$, where i is the number of the energy level, (j) is the order of "our" perturbation theory. Thus, e.g., $E_0^{(4)}$ is the 4th order correction of the 0-th state. We also use the convention $\bar{E}_i^{(2)} = E_i^{(0)} + E_i^{(2)}$, $\bar{E}_i^{(4)} = E_i^{(0)} + E_i^{(2)} + E_i^{(4)}$, $R_i^{(j)} = \bar{E}_i^{(j)}/E_i^{(e)}$, where $E_i^{(e)}$ are the exact values of ref. 1.

§7 contains a discussion of (4.4) for the large λ regime for $i \geqslant 2$, ($i = 0,1$ have already been discussed in §5). The results are (7.3) and (7.4), whose numerical accuracies are discussed in §9. §8 contains a discussion of (4.4) for $\lambda \to \infty$, with finite large and small $\beta = i\lambda$, and the results are our eq. (8.3), which covers any β values quite well, contrary to (1-1.18b) which is valid only for small β. §9 gives some numerical results. For the states i=0,1, our fourth approximation results based on (5.12) and (5.17) are summarized in Tables 1 and 2, for various intermediate λ. We see that our formulae give excellent results also for such an intermediate regime of λ. For states $i \geqslant 2$, our numerical estimate for intermediate values of λ is based on our second approximation formula (4.4). The errors, as compared with the exact values, can be most clearly seen in Fig.4. We summarize for i=2,3,8 the numerical results in Table 4. The i=8 state is chosen as an example, for an intermediate i. As we will see below, almost all our calculations are algebraic, we never used integrations here, and only once differentiated in eq. (2.7a). We reduced a difficult nonlinear differential equation's eigenvalue problem to an algebraic manipulation. Our most difficult task here is to solve the cubic equation (2.10). The calculations are really elementary, although for higher approximations the arithmetic becomes somewhat complicated. The numerical calculations are also very simple and almost all the numerical computation was done by the author on a desk computer.

Next we should mention something about the symmetry properties. As our total Hamiltonian has even parity (even function of p and q), any sensible approximation should

not destroy this parity property. This means, in our case, that the state with i = even (odd) should be an even (odd) state. Our approximation retains, of course, this symmetry property. So the i=1 state is the lowest state of the i=odd states, which is guaranteed to be orthogonal to the i=0 state with even parity. This means, it is guaranteed that $E_0^{(o)} \geqslant E_0^{(e)}$ and $E_1^{(o)} \geqslant E_1^{(e)}$ due to the variational character of our zeroth order approximation, but for other i and for higher approximations we have no guarantee whether our calculated values are larger or smaller than the exact values. [We will encounter this problem, for example, for i=8. We see from Fig.4 that $R_8^{(0)} < 1$ and $R_8^{(2)} > 1$.]

Lastly we want to discuss our wave functions. The lowest order wave function, corresponding to the eigenvalue $E_i^{(0)}$ is given by (3.3). It is just the usual Hermite function, only scaled appropriately by our α_i. We have not given the corresponding wave function for $E_i^{(2)}$, but it is quite easily constructed explicitly, if necessary, by a standard perturbation procedure using $H_i^{(I)}$ as perturbation. Our states are not orthogonal to each other, because they are based on differently scaled Hermite functions.

§ 2. The Method of Approximation

Our anharmonic oscillator is characterized by the Hamiltonian

$$H = \frac{1}{2}(p^2 + \omega^2 q^2) + \lambda' q^4 \tag{2.1}$$

We have chosen units in which $\hbar = m = 1$.
Let us substitute p and q by

$$\left.\begin{aligned} &q = \sqrt{\frac{\alpha}{2\omega}}\,(b + b^+), \quad p = \frac{1}{i}\sqrt{\frac{\omega}{2\alpha}}\,(b - b^+) \\ &[b, b^+] = 1 \end{aligned}\right\} \tag{2.2}$$

where α is our characteristic variational parameter, which will play the central role in our approximation method and will be defined below. $\alpha=1$ leads to the well known creation and annihilation operators a and a^+ of the simple harmonic oscillator. Inserting (2.2) into our Hamiltonian (2.1) and expressing H in terms of b and b^+, we then arrange this into Wick's normal product ordering to get easily

$$H = H^{(O)} + H^{(I)}$$

$$\left.\begin{aligned} \frac{H^{(O)}}{\omega} &= \left(\frac{1}{4\alpha} + \frac{\alpha}{4} + \frac{3\lambda\alpha^2}{4}\right)(2n+1) + \frac{3\lambda\alpha^2}{2} n^2 \\ \frac{H^{(I)}}{\omega} &= \left(-\frac{1}{4\alpha} + \frac{\alpha}{4} + \frac{3\lambda\alpha^2}{2}\right)(b^2 + b^{+2}) \\ &\quad + \frac{\lambda\alpha^2}{4}(b^4 + 4nb^2 + 4b^{+2}n + b^{+4}) \end{aligned}\right\} \tag{2.3}$$

where

$$n \equiv b^+ b, \qquad \lambda \equiv \frac{\lambda'}{\omega^3} \tag{2.4}$$

$H^{(0)}$ is the diagonal part and $H^{(I)}$ is the nondiagonal part of the total H, with respect to n. We define as usual

$$\left.\begin{aligned} |i\rangle &\equiv \frac{1}{\sqrt{i!}}(b^{+})^{i}|0\rangle \quad , (i = 1, 2, \cdots) \\ b|0\rangle &= 0 \end{aligned}\right\} \tag{2.5}$$

and get

$$\left.\begin{aligned} &H^{(0)}|i\rangle = E_i^{(0)}(\alpha)|i\rangle \\ &\text{and} \\ &\langle i|H^{(I)}|i\rangle = 0 \end{aligned}\right\} \tag{2.6}$$

where

$$\frac{E_i^{(0)}(\alpha)}{\omega} = \left(\frac{1}{4\alpha} + \frac{\alpha}{4} + \frac{3\lambda\alpha^2}{4}\right)(2i+1) + \frac{3\lambda\alpha^2}{2} i^2 \tag{2.7}$$ *)

and then

$$\frac{\partial E_i^{(0)}(\alpha)}{\partial \alpha} = \left(-\frac{1}{4\alpha^2} + \frac{1}{4} + \frac{3\lambda\alpha}{2}\right)(2i+1) + 3\lambda\alpha\, i^2 \tag{2.7a}$$

Now we choose our α to optimize $E_i^{(0)}(\alpha)$, namely we define our α_i by

$$\frac{\partial E_i^{(0)}(\alpha)}{\partial \alpha} = 0 \longrightarrow \alpha = \alpha_i$$

so

$$\alpha_i^3\left(6\lambda\cdot\frac{2i^2+2i+1}{2i+1}\right) + \alpha_i^2 - 1 = 0 \tag{2.8}$$

+) From here on we put $\omega=1$, without loss of generality.

For states $i \geq 2$ this type of variational determination of α_i may be questionable. We can justify our choice of α_i in the following way: For our Hamiltonian H, the virial theorem requires

$$\langle \tfrac{1}{2} p^2 \rangle = \langle \tfrac{1}{2} q^2 + 2\lambda q^4 \rangle$$

for any eigenstate of H. If we require this to be valid also for our approximate eigenstates $|i\rangle$, the above equation becomes

$$\frac{1}{4\alpha}(2i+1) = \frac{\alpha}{4}(2i+1) + \frac{3\lambda\alpha^2}{2}(2i^2+2i+1)$$

which is nothing else than eq. (2.8).

Now define

$$\left.\begin{aligned} y_i &\equiv 6\lambda\cdot\frac{(2i^2+2i+1)}{(2i+1)} \equiv 6\bar{\Lambda}_i \\ \bar{\Lambda}_i &\equiv \lambda\cdot\frac{(2i^2+2i+1)}{(2i+1)} = (i+\tfrac{1}{2})\lambda\left\{1+\frac{1}{4(i+\frac{1}{2})^2}\right\} \\ &= \Lambda_i\left\{1+\frac{1}{4(i+\frac{1}{2})^2}\right\} \\ \Lambda_i &\equiv (i+\tfrac{1}{2})\lambda \end{aligned}\right\} \qquad (2.9)$$

Λ_i is the notation used in ref.1. With this y_i our α_i is given by a solution of

$$y_i\alpha_i^3 + \alpha_i^2 - 1 = 0 \qquad (2.10)$$

This equation determines our variational parameter α_i in terms of y_i, which is a special combination of λ and i (the level number). y_i is essentially $6\Lambda_i$ of reference 1, except for the factor $1+\frac{1}{4(i+\frac{1}{2})^2}$, which plays some role for small i, but is practically one for large i. <u>In our approximation we choose α_i for each level i differently</u>. For fixed i we have only one special expansion parameter y_i, which can be small or large, corresponding to small or large λ, and our lowest approximation result $E_i^{(0)}$ depends only on α_i (eq. (2.11), see below) which covers all ranges of λ.

With this α_i our $E_i^{(0)}$ becomes

$$\left.\begin{aligned} E_i^{(0)}(\alpha_i) &= \frac{2i+1}{8}\left(\frac{3}{\alpha_i}+\alpha_i\right) \\ \text{or}\qquad \varepsilon_i^{(0)} &\equiv \frac{E_i^{(0)}(\alpha)}{i+\frac{1}{2}} = \frac{1}{4}\left(\frac{3}{\alpha_i}+\alpha_i\right) \end{aligned}\right\} \qquad (2.11)$$

where α_i is defined as a real positive solution of the cubic eq. (2.10). Eq. (2.10) has three real solutions for $27y_i^2<4$. In this case we choose the positive root, which leads to $\alpha_i \to 1$ for $\lambda \to 0$, the correct usual perturbation calculation.

In summarizing up to here, we have split our total H into a nonperturbed part $H_i^{(0)}$ and a perturbed part $H_i^{(I)}$, which are given by

$$\left.\begin{aligned} H_i^{(0)} &= \left(\frac{1}{4\alpha_i} + \frac{\alpha_i}{4} + \frac{3\lambda\alpha_i^2}{4}\right)(2n_i+1) + \frac{3\lambda\alpha_i^2}{2} n_i^2 \\ H_i^{(I)} &= \left(-\frac{1}{4\alpha_i} + \frac{\alpha_i}{4} + \frac{3\lambda\alpha_i^2}{2}\right)(b_i^2 + b_i^{+2}) \\ &\quad + \lambda\alpha_i^2(n_i b_i^2 + b_i^{+2} n_i) + \frac{\lambda\alpha_i^2}{4}(b_i^4 + b_i^{+4}) \end{aligned}\right\} \tag{2.12}$$

where $n_i = b_i + b_i$ is still <u>an operator</u>. Our $H_i^{(0)}$ is diagonal but nonlinear (containing n_i^2). We separate our total H into different parts $H_i^{(0)}$ and $H_i^{(I)}$ for each level i. (α_i is different for different i, defined by eq. (2.10)).
Our b_i and b_i^+ are different for each level, and are given by

$$\begin{aligned} b_i &= \sqrt{\frac{\omega}{2\alpha_i}}\, q + i\sqrt{\frac{\alpha_i}{2\omega}}\, p \\ &= \frac{1}{2}\left\{\left(\frac{1}{\sqrt{\alpha_i}} + \sqrt{\alpha_i}\right) a + \left(\frac{1}{\sqrt{\alpha_i}} - \sqrt{\alpha_i}\right) a^+\right\} \end{aligned} \tag{2.13}$$

where a and a^+ are the usual creation and annihilation operators for the free harmonic oscillator. b_i^+ is the hermitean conjugate of b_i. Thus $[b_i, b_j] \neq 0$ for $i \neq j$. Looking at (2.13), our method amounts to performing different Bogoliubov transformations for each i and λ*).

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*) I am indebted to H.Mitter for pointing out this fact.

In our approximation method we attack each level separately. For fixed i, α_i is still adjusted for different λ. Thus choosing our best α_i as a solution of eq. (2.10), our method amounts to performing a perturbation procedure by treating $H_i^{(0)}$ as the unperturbed part and $H_i^{(I)}$ as perturbation. As we will see below, $H_i^{(I)}$ is always small compared to $H_i^{(0)}$, but does <u>not</u> contain any small parameter of fixed value, so that our method is valid for all ranges of $0<\lambda<\infty$ and $0\leq i<\infty$, with one <u>simple analytic formula</u>.

§3. The Lowest Order Approximation for Arbitrary λ and i.

The lowest order approximation is to neglect $H_i^{(I)}$ and retain only $H_i^{(0)}$. Then the results were already given in §2, namely

$$\varepsilon_i^{(0)} \equiv \frac{E_i^{(0)}}{i+\frac{1}{2}} = \frac{1}{4}\left(\frac{3}{\alpha_i} + \alpha_i\right) \tag{3.1}$$

where α_i is the positive real root of

$$\left.\begin{aligned} & y_i \alpha_i^3 + \alpha_i^2 - 1 = 0 \\ \text{where} \quad & y_i = 6\lambda\left(\frac{2i^2+2i+1}{2i+1}\right) \end{aligned}\right\} \tag{3.2}$$

We give $\varepsilon_i^{(0)}$ and α_i as a function of y_i in Figs. 1,2 and 3. Fig. 2 demonstrates the accuracy of our lowest order result $\varepsilon_i^{(0)}$ for the small y_i region.

If we calculate energy levels for i=0, 1,2 ..... 8 as functions of λ from eq. (3.1), and plot a curve, then we get quite a similar graph as given in (1-Fig. 1), and we cannot see at all any difference at this scale of graphs. In order to see the errors of our approximation we give a graph of the ratio $R_i^{(0)} = E_i^{(0)}/E_i^{(e)}$ (i=0,1,2,3 and 8) as a function of λ in Fig. 4, where $E_i^{(0)}$ are our calculated values obtained from eq. (3.1),and $E_i^{(e)}$ are the exact values given by ref. 2. The upper suffix (0) means the lowest approximation values. In Fig. 4 we also give the values obtained by our second approximation shown by a dotted curve $R_i^{(2)}$ with suffix (2). As we will see from Fig. 4, our $R_i^{(0)}$ has at most an error of 2%

(for i=0, 1; $\lambda \gg 1$) and the error is usually less than 1% ($0.99 < R_i^{(0)} < 1.01$) for almost every value of i and λ. (As we will see in §8, this is true even for $i \gg 1$, namely in the WKB region).
As our approach is variational, it is guaranteed that $R_0^{(0)}$ and $R_1^{(0)}$ are larger than 1, while the $R_i^{(0)}$ ($i \geq 2$) are not necessarily larger than 1, because our excited levels are not orthogonal to the lower levels. The fact that $R_1^{(0)}$ is larger than 1 comes from the symmetry property of our total H: Since H is an even function of q and p, the ground state is an even function while the first excited state is an odd function, and our approximation does does not destroy this symmetry.

The corresponding states are given by

$$|i^{(0)}\rangle = \frac{1}{\sqrt{i!}}(b_i^+)^i |0\rangle$$

Thus the wave functions in the q-representation are the usual Hermite polynomials, but appropriately scaled by α_i:

$$\psi_i^{(0)}(q) \equiv \langle q | i^{(0)} \rangle = \left(\frac{\omega}{\pi \alpha_i}\right)^{\frac{1}{4}} \frac{1}{\sqrt{2^i i!}} H_i\left(\sqrt{\frac{\omega}{\alpha_i}}\, q\right) e^{-\frac{\omega}{2\alpha_i} q^2} \tag{3.3}$$

§4. The Second Order Perturbation due to $H_i^{(I)}$ for Arbitrary λ and i.

The first order perturbation due to $H_i^{(I)}$ is zero, as was shown in (2.6) : $\langle i|H_i^{(I)}|i\rangle = 0$.

The second order perturbation is given by

$$
\begin{aligned}
E_i^{(2)} &= \langle i|\, H_i^{(I)} \frac{1}{E_i^{(0)} - H_i^{(0)}(n)}\, H_i^{(I)}\, |i\rangle \\
&= \frac{1}{i!}\left(\frac{\lambda\alpha_i^2}{4}\right)^2 \Bigg[\langle 0|\, b^{i+4} \frac{1}{E_i^{(0)} - H_i^{(0)}(n=i+4)} (b^+)^{i+4}\, |0\rangle \\
&\qquad + \langle 0|\, b^{i} (b^+)^{4} \frac{1}{E_i^{(0)} - H_i^{(0)}(n=i-4)}\, b^4 (b^+)^{i}\, |0\rangle \Bigg] \\
&+ \frac{1}{i!}\left(-\frac{1}{4\alpha_i} + \frac{\alpha_i}{4} + \frac{3\lambda\alpha_i^2}{2} + i\lambda\alpha_i^2\right)^2 \langle 0|\, b^{i+2} \frac{1}{E_i^{(0)} - H_i^{(0)}(n=i+2)} (b^+)^{i+2}\, |0\rangle \\
&+ \frac{1}{i!}\left(-\frac{1}{4\alpha_i} + \frac{\alpha_i}{4} + \frac{3\lambda\alpha_i^2}{2} + (i-2)\lambda\alpha_i^2\right)^2 \langle 0|\, b^{i} (b^+)^{2} \frac{1}{E_i^{(0)} - H_i^{(0)}(n=i-2)}\, b^2 (b^+)^{i}\, |0\rangle \\
&= \left(\frac{\lambda\alpha_i^2}{4}\right)^2 \left[\frac{(i+4)(i+3)(i+2)(i+1)}{E_i^{(0)} - H_i^{(0)}(n=i+4)} + \frac{i(i-1)(i-2)(i-3)}{E_i^{(0)} - H_i^{(0)}(n=i-4)} \right] \\
&+ \left(\frac{1}{4\alpha_i}\right)^2 \Bigg[(i+2)(i+1)\left\{-1 + \alpha_i^2 + 2\lambda\alpha_i^3(2i+3)\right\}^2 \frac{1}{E_i^{(0)} - H_i^{(0)}(n=i+2)} \\
&\qquad + i(i-1)\left\{-1 + \alpha_i^2 + 2\lambda\alpha_i^3(2i-1)\right\}^2 \frac{1}{E_i^{(0)} - H_i^{(0)}(n=i-2)} \Bigg]
\end{aligned}
\tag{4.1}
$$

Using the relation for α_i given by (2.10), and y_i given in eq. (2.9), we obtain after some elementary algebra

$$E_i^{(2)} = \left(\frac{1-\alpha_i^2}{\alpha_i} \cdot \frac{1}{6(2i^2+2i+1)}\right)^2$$

$$\times \left[\left(\frac{2i+1}{4}\right)^2 \left\{\frac{(i+4)(i+3)(i+2)(i+1)}{E_i^{(0)} - H_i^{(0)}(n=i+4)} + \frac{i(i-1)(i-2)(i-3)}{E_i^{(0)} - H_i^{(0)}(n=i-4)}\right\}\right. \tag{4.2}$$

$$\left. + (i+2)(i+1)i(i-1)\left\{\frac{i(i-1)}{E_i^{(0)} - H_i^{(0)}(n=i+2)} + \frac{(i+2)(i+1)}{E_i^{(0)} - H_i^{(0)}(n=i-2)}\right\}\right]$$

while

$$\begin{aligned} V_i(k) &\equiv E_i^{(0)} - H_i^{(0)}(n=i+k) \\ &= -\left(\frac{1}{4\alpha_i} + \frac{\alpha_i}{4} + \frac{3\lambda\alpha_i^2}{4}\right)(2k) - \frac{3\lambda\alpha_i^2}{2}\left\{(i+k)^2 - i^2\right\} \\ &= -\frac{k}{4\alpha_i} \cdot \frac{1}{(2i^2+2i+1)}\left[8i^2 + 2i(k+4) + (k+3) - (2ki+k-1)\alpha_i^2\right] \end{aligned} \tag{4.3}$$

(k can be negative also)

and we finally get

$$E_i^{(2)} = \left(\frac{1-\alpha_i^2}{6}\right)^2 \frac{1}{\alpha_i(2i^2+2i+1)} \tag{4.4}$$

$$\times \left[\left(\frac{2i+1}{4}\right)^2 \left\{-\frac{(i+4)(i+3)(i+2)(i+1)}{8i^2+16i+7-\alpha_i^2(8i+3)} + \frac{i(i-1)(i-2)(i-3)}{8i^2-1+\alpha_i^2(8i+5)}\right\}\right.$$

$$\left. + 2(i+2)(i+1)i(i-1)\left\{-\frac{i(i-1)}{8i^2+12i+5-\alpha_i^2(4i+1)} + \frac{(i+2)(i+1)}{8i^2+4i+1+\alpha_i^2(4i+3)}\right\}\right]$$

Thus our second order perturbation results are given by $\bar{E}_i^{(2)} = E_i^{(0)}+E_i^{(2)}$, where $E_i^{(0)}$ is given by (3.1). This expression has some resemblance to (1-1.17), because they are related by a similar calculational way. But they are not the same, and above all (1-1.17) is valid only for small λ, while our results are valid for arbitrary λ. The two expression deviate at about $\beta\sim 1$, as will be clearly seen from the graph of Fig.4.

As we see from the graphs of Fig. 4, $\left|(\bar{E}_i^{(2)}-E_i^{(e)})/E_i^{(e)}\right|$ are usually less than 0.01 for all ranges of λ and i, and furthermore $E_i^{(2)}$ has the correct sign and magnitude to reduce the error of $E_i^{(0)}$ by a factor of about 1/5.

As we have already remarked in §3, we always have $R_0^{(0)}\geq 1$ and $R_1^{(0)}\geq 1$, while,as we may easily see from (4.4), if we put i=0 or i=1 there, $E_0^{(2)}<0$ and $E_1^{(2)}<0$, so our second approximation gives $R_i^{(0)} \equiv E_i^{(0)}/E_i^{(e)} > \bar{E}_i^{(2)}/E_i^{(e)} \equiv R_i^{(2)}$ for i=0,1. But from the beginning it is not guaranteed that $R_i^{(2)}\geq 1$ for i=0,1; only after numerical calculations we found this to be the case.

Similarly for $i\geq 2$, generally speaking we are not sure whether $R_i^{(0)}$ is larger or smaller than 1, and $R_i^{(2)}$ is nearer to 1 than $R_i^{(0)}$. Only after numerical calculations we see that this is really the case for all ranges of λ and i (even for $i\to\infty$).

§5. The Fourth Order Perturbation due to $H_i^{(I)}$, for the Ground (i=0) and First Excited (i=1) States for arbitrary λ.

We may continue "our" perturbation calculation due to $H_i^{(I)}$ systematically to higher orders, but in full generality this would become much more complicated and the numerical improvements are expected to be quite small, which will be shown below to be true. So we will concentrate in this section only on the ground and the first excited states (i=0,1), but still leave λ arbitrary.

The calculations of states with i=0,1 are especially simple, as can be seen in formula (4.4) of the previous section. In (4.4), for i=0,1, only the first term of the four terms in [. .....] survives, which originates from b^4 and b^{+4} in $H_o^{(I)}$ and $H_1^{(I)}$. This is due to our choice of α_o and α_1, namely

$$\left(-\frac{1}{4\alpha_o}+\frac{\alpha_o}{4}+\frac{3\lambda\alpha_o^2}{2}\right)=0 \qquad \text{for any } \lambda,$$

since $y_o = 6\lambda$ (see eq. (2.9)). Then the coefficient of the $(b_o^2+b_o^{+2})$ - term of $H_o^{(I)}$ is zero. Hence our $H_o^{(I)}$ becomes simply

$$\left.\begin{aligned} H_o^{(I)} &= \frac{\lambda\alpha_o^2}{4}\left(b_o^4+4n_ob_o^2+4b_o^{+2}n_o+b_o^{+4}\right) \\ &= \left(\frac{1-\alpha_o^2}{24\alpha_o}\right)\left(\qquad '' \qquad \right) \end{aligned}\right\} \qquad (5.1)$$

Hence clearly

$$\left.\begin{aligned} & H_0^{(I)}|0\rangle = \frac{\lambda\alpha_0^2}{4} b_0^{+4}|0\rangle \sim |4\rangle \\ \text{i.e} \quad & \langle 2|H_0^{(I)}|0\rangle = \langle 0|H_0^{(I)}|2\rangle = 0 \qquad \text{for any } \lambda \end{aligned}\right\} \quad (5.2)$$

identically. This makes our calculation for the ground state (i=0) especially simple.

For the i=1 state we have $y_1 = 10\lambda$ so we have

$$\left(-\frac{1}{4\alpha_1} + \frac{\alpha_1}{4} + \frac{3\lambda\alpha_1^2}{2}\right) = -\lambda\alpha_1^2$$

and

$$H_1^{(I)} = -\lambda\alpha_1^2(b_1^2 + b_1^{+2}) + \frac{\lambda\alpha_1^2}{4}(b_1^4 + 4n_1 b_1^2 + 4b_1^{+2}n_1 + b_1^{+4})$$

This time the coefficient of $(b_1^2+b_1^{+2})$ does not vanish, but we can write

$$\left.\begin{aligned} H_1^{(I)} &= \left(\frac{\lambda\alpha_1^2}{4}\right)\left[b_1^4 + (b_1^+)^4 + 4(n_1-1)b_1^2 + 4b_1^{+2}(n_1-1)\right] \\ &= \left(\frac{1-\alpha_1^2}{40\alpha_1}\right)\left[\qquad '' \qquad \right] \end{aligned}\right\} \quad (5.3)$$

And we see

$$b_1^{+2}(n_1-1)|1\rangle = 0$$

so that

$$\left.\begin{aligned} & H_1^{(I)}|1\rangle = \left(\frac{\lambda\alpha_1^2}{4}\right)(b_1^+)^4|1\rangle \sim |5\rangle \\ \therefore \quad & \langle 3|H_1^{(I)}|1\rangle = \langle 1|H_1^{(I)}|3\rangle = 0 \qquad \text{for any } \lambda \end{aligned}\right\} \quad (5.4)$$

The first excited state, i=1, is also simple, which is the lowest state with odd parity.

For states $i \geq 2$, the situation is much more complicated as can be seen already from the second order calculation of the last section and formula(4.4).

The properties (5.2) and (5.4) guarantee that the third order perturbations are zero, since

$$\left.\begin{aligned} &\langle 0| H_0^{(I)} H_0^{(I)} H_0^{(I)} |0\rangle \sim \langle 4| H_0^{(I)} |4\rangle = 0 \\ &\langle 1| H_1^{(I)} H_1^{(I)} H_1^{(I)} |1\rangle \sim \langle 5| H_1^{(I)} |5\rangle = 0 \end{aligned}\right\} \qquad (5.5)$$

On the other hand this is generally not the case for $i \geq 2$, as for example

$$\langle 2| H_2^{(I)} H_2^{(I)} H_2^{(I)} |2\rangle \sim \langle 2| H_2^{(I)} H_2^{(I)} |0\rangle \sim \langle 2| H_2^{(I)} |4\rangle \neq 0 \qquad (5.6)$$

For the same reasoning, for the states i=0,1 the fifth order perturbations are not zero, as for example

$$\langle 0| H_0^{(I)} H_0^{(I)} H_0^{(I)} H_0^{(I)} H_0^{(I)} |0\rangle \sim \langle 4| H_0^{(I)} H_0^{(I)} H_0^{(I)} |4\rangle$$

$$\sim \langle 8| H_0^{(I)} |6\rangle \neq 0$$

since $H_0^{(I)}$ contains $b_0^{+2} n_0$, which brings $|4\rangle \longrightarrow |6\rangle$.

With these preliminary remarks let us now begin our fourth order perturbation calculation for i=0 and i=1. For i=0,1 we have

$$H_0^{(0)}(n_0) = \frac{1}{4\alpha_0}\left\{(3+\alpha_0^2)(n_0+\tfrac{1}{2}) + (1-\alpha_0^2)\,n_0^2\right\} \tag{5.7}$$

$$H_1^{(0)}(n_1) = \frac{1}{20\,\alpha_1}\left\{(13+7\alpha_1^2)(n_1+\tfrac{1}{2}) + 3\,(1-\alpha_1^2)\,n_1^2\right\} \tag{5.8}$$

And, as in eq. (4.3),

$$V_i(k) \equiv E_i^{(0)} - H_i^{(0)}(n=i+k) = H_i^{(0)}(n=i) - H_i^{(0)}(n=i+k)$$

$$V_0(k) = -\frac{k}{4\alpha_0}\left\{3 + k - \alpha_0^2(k-1)\right\} \tag{5.9}$$

$$V_1(k) = -\frac{k}{20\alpha_1}\left\{19 + 3k - \alpha_1^2(3k-1)\right\} \tag{5.10}$$

We will explicitly carry out the calculation but for i=1 we will state only the results.

The standard perturbation theory gieves us

$$E_0^{(4)} = \langle 0|H_0^{(I)}\frac{1}{E_0^{(0)}-H_0^{(0)}}\left\{H_0^{(I)}\frac{P}{E_0^{(0)}-H_0^{(0)}}H_0^{(I)} - \langle 0|H_0^{(I)}\frac{1}{E_0^{(0)}-H_0^{(0)}}H_0^{(I)}|0\rangle\right\}\frac{1}{E_0^{(0)}-H_0^{(0)}}H_0^{(I)}|0\rangle$$

where

$$P \equiv 1 - |0\rangle\langle 0|$$

is the projection operator onto the non - |0> states. From (5.2) we get

$$E_0^{(4)} = \left(\langle 0 | H_0^{(I)} | 4 \rangle \frac{1}{V_0(4)} \right)^2 \left\{ \langle 4 | H_0^{(I)} \frac{P}{E_0^{(0)} - H_0^{(0)}} H_0^{(I)} | 4 \rangle - E_0^{(2)} \right\}$$

$$= |\langle 0 | H_0^{(I)} | 4 \rangle|^2 \left(\frac{1}{V_0(4)} \right)^2 \left\{ \sum_{k=2,6,8} |\langle 4 | H_0^{(I)} | k \rangle|^2 \frac{1}{V_0(k)} - E_0^{(2)} \right\} \tag{5.11}$$

where $E_0^{(2)}$ is given by (4.4) with i=0, namely

$$E_0^{(2)} = |\langle 0 | H_0^{(I)} | 4 \rangle|^2 \frac{1}{V_0(4)} \quad ,$$

in this notation. $V_0(k)$ is given in (5.9) and

$$\langle 0 | H_0^{(I)} | 4 \rangle = \frac{1}{\sqrt{4!}} \left(\frac{1-\alpha_0^2}{24\alpha_0} \right) \langle 0 | b_0^4 \, b_0^{+4} | 0 \rangle = \sqrt{4!} \left(\frac{1-\alpha_0^2}{24\alpha_0} \right) \qquad \text{etc.}$$

So after some calculation we arrive at

$$E_0^{(4)} = \left(\frac{1-\alpha_0^2}{24\alpha_0} \right)^4 (-4\alpha_0)^3 \left(\frac{1}{4(7-3\alpha_0^2)} \right)^2$$

$$\times \left[\frac{(4\cdot 4\cdot 3\cdot 2)^2 \, 2!}{2(5-\alpha_0^2)} + \frac{(4\cdot 4)^2 \, 6!}{6\,(9-5\alpha_0^2)} + \frac{8!}{8\,(11-7\alpha_0^2)} - \frac{(4!)^2}{4(7-3\alpha_0^2)} \right]$$

$$= -\frac{(1-\alpha_0^2)^4}{72\alpha_0} \frac{1}{(7-3\alpha_0^2)^2} \left[\frac{8}{5-\alpha_0^2} + \frac{80}{3(9-5\alpha_0^2)} + \frac{35}{8(11-7\alpha_0^2)} - \frac{1}{8(7-3\alpha_0^2)} \right]$$

So our fourth order final result for i=0 and arbitrary λ is given by

$$\bar{E}_0^{(4)} = E_0^{(0)} + E_0^{(2)} + E_0^{(4)}$$

where

$$\left.\begin{aligned}
E_0^{(0)} &= \frac{1}{8}\left(\frac{3}{\alpha_0} + \alpha_0\right) \\
E_0^{(2)} &= -\frac{(1-\alpha_0^2)^2}{24\alpha_0}\,\frac{1}{7-3\alpha_0^2} \\
E_0^{(4)} &= -\frac{(1-\alpha_0^2)^4}{72\alpha_0}\,\frac{1}{(7-3\alpha_0^2)^2} \\
&\quad\times\left[\frac{8}{5-\alpha_0^2} + \frac{80}{3(9-5\alpha_0^2)} + \frac{35}{8(11-7\alpha_0^2)} - \frac{1}{8(7-3\alpha_0^2)}\right] \\
&\text{and } \alpha_0 \text{ is the positive real root of} \\
&\quad 6\lambda\alpha_0^3 + \alpha_0^3 - 1 = 0.
\end{aligned}\right\} \quad (5.12)$$

This formula covers all ranges of λ, and the worst case $\lambda \to \infty$, gives only a 0.01% error compared to the exact numerical results obtained in ref. 2. As can be seen from our Fig.4, if we tolerated a maximum error of 0.4%, then the much simpler formula $E_0^{(0)}+E_0^{(2)}$ of §4 is sufficient for our purposes. And even our extremely simple formula $E_0^{(0)}$ covers all ranges of λ within a maximum error of 2%, as we can clearly see from Fig. 4. A more detailed numerical comparison in the intermediate λ region will be given in Table 1 of the final section §9.

Let us first discuss the case $\lambda >> 1$: We get for the leading terms

$$E_0^{(0)} = \frac{3}{8}(6)^{\frac{1}{3}} \lambda^{\frac{1}{3}} + \frac{1}{4}(6)^{-\frac{1}{3}} \lambda^{-\frac{1}{3}} + \cdots\cdots$$

$$E_0^{(2)} = \frac{3}{8}(6)^{\frac{1}{3}} \left(-\frac{1}{63}\right) \lambda^{\frac{1}{3}} + \frac{1}{4}(6)^{-\frac{1}{3}} \left(+\frac{13}{336}\right) \lambda^{-\frac{1}{3}} + \cdots\cdots$$

$$E_0^{(4)} = \frac{3}{8}(6)^{\frac{1}{3}} \left\{ -\frac{1}{3969}\left(\frac{24}{5} + \frac{80}{9} + \frac{105}{88}\right) + \frac{1}{74088} \right\} \lambda^{\frac{1}{3}} \tag{5.13}$$

$$+ \frac{1}{4}(6)^{-\frac{1}{3}} \left[\frac{1}{54} \cdot \frac{1}{49} \left\{ \frac{59}{21}\left(\frac{24}{5} + \frac{80}{9} + \frac{105}{88}\right) - \left(\frac{24}{25} + \frac{400}{88} + \frac{735}{968}\right) \right\} + \frac{17}{518616} \right] \lambda^{-\frac{1}{3}} + \cdots\cdots$$

and so we finally obtain

$$\overline{E}_4^{(0)} = E_0^{(0)} + E_0^{(2)} + E_0^{(4)} = 0.6680582\, \lambda^{\frac{1}{3}} + 0.143468\, \lambda^{-\frac{1}{3}} + \cdots\cdots \tag{5.13a}$$

This should be compared to the exact values (cf. 1-Table1)

$$E_0^{(e)} = 0.6679826\, \lambda^{\frac{1}{3}} + 0.14367\, \lambda^{-\frac{1}{3}} + \cdots\cdots$$

Next let us discuss the case of λ being small: For λ small our α_0 is given by

$$\alpha_0 = 1 - 3\lambda + \frac{5}{2}(3\lambda)^2 - 8(3\lambda)^3 + \frac{231}{8}(3\lambda)^4 + \cdots\cdots \tag{5.14}$$

and our $H_o^{(I)}$ contains as a factor $\lambda\alpha_o^2$, (see (5.1)). If we perform the ℓ-th order of "our" perturbation theory, we neglect only terms of order $\lambda^{\ell+1}$ so that our formula should give the correct usual perturbation result up to order λ^{ℓ}. This means, our $E_o^{(0)}$ gives the correct λ^0 and λ^1 coefficients (because $E_o^{(1)}=0$), $E_o^{(0)}+E_o^{(2)}$ goes up to λ^3 (because $E_o^{(3)}=0$), $E_o^{(0)}+E_o^{(2)}+E_o^{(4)}$ gives the correct coefficients up to λ^4 (because $E_o^{(5)}\neq 0$). We will show that this is really the case, and this fact will also guarantee that we are on the right track. If we systematically expand our formulae for small λ, using (5.14), we get

$$\left.\begin{aligned}
E_o^{(0)} &= \frac{1}{2} + \frac{1}{4}(3\lambda) - \frac{1}{4}(3\lambda)^2 + \frac{1}{2}(3\lambda)^3 - \frac{21}{16}(3\lambda)^4 + \cdots\cdots \\
E_o^{(2)} &= \qquad\qquad - \frac{1}{24}(3\lambda)^2 + \frac{13}{48}(3\lambda)^3 - \frac{49}{32}(3\lambda)^4 + \cdots\cdots \\
E_o^{(4)} &= \qquad\qquad\qquad\qquad - \frac{467}{3456}(3\lambda)^4 + \cdots\cdots
\end{aligned}\right\} \quad (5.15)$$

$$\therefore \quad \left.\begin{aligned}
E_o^{(0)} &= \frac{1}{2} + \frac{3}{4}\lambda + \cdots\cdots \\
E_o^{(0)} + E_o^{(2)} &= \frac{1}{2} + \frac{3}{4}\lambda - \frac{21}{8}\lambda^2 + \frac{333}{16}\lambda^3 + \cdots\cdots \\
E_o^{(0)} + E_o^{(2)} + E_o^{(4)} &= \frac{1}{2} + \frac{3}{4}\lambda - \frac{21}{8}\lambda^2 + \frac{333}{16}\lambda^3 - \frac{30885}{128}\lambda^4 + \cdots
\end{aligned}\right\} \quad (5.16)$$

which reproduce the well known correct numerical coefficients given for example by Bender and Wu[3] up to λ, λ^3 and λ^4, respectively, as was argued above. This means, our formulae cover also the weak coupling ($\lambda \ll 1$) region quite satisfactorily, as it should be. But our formula (5.12) covers <u>all ranges of λ</u> with quite small error. As was already remarked above, since $E_o^{(5)}$ is not zero, we cannot get the correct λ^5 coefficient if we expand (5.12). The fact that our $E_o^{(0)}+E_o^{(2)}$ gives the correct λ^3-coefficient is due to the special circumstance that $E_o^{(3)}=0$.

Now, for the first excited state (i=1), we have instead of (5.12):

$$\left.\begin{aligned}
\bar{E}_1^{(4)} &= E_1^{(0)} + E_1^{(2)} + E_1^{(4)} \\
&\text{where} \\
E_1^{(0)} &= \frac{3}{8}\left(\frac{3}{\alpha_1} + \alpha_1\right) \\
E_1^{(2)} &= -\frac{3(1-\alpha_1^2)^2}{8\alpha_1} \cdot \frac{1}{31-11\alpha_1^2} \\
E_1^{(4)} &= -\frac{(1-\alpha_1^2)^4}{\alpha_1} \frac{3}{(31-11\alpha_1^2)^2} \\
&\quad \times \left[\frac{1}{5-\alpha_1^2} + \frac{14}{37-17\alpha_1^2} + \frac{189}{64(43-23\alpha_1^2)} - \frac{1}{512(31-11\alpha_1^2)}\right] \\
&\text{and } \alpha_i \text{ is the positive real root of} \\
&10\lambda\alpha_1^3 + \alpha_1^2 - 1 = 0
\end{aligned}\right\} \quad (5.17)$$

This formula covers all ranges of λ, and like $\bar{E}_o^4$, even for $\lambda\to\infty$ the error is 0.013%, while $E_o^{(0)}+E_o^{(2)}$ has at most an error of 0.17%, and the simplest expression $E_1^{(0)}$ gives an error of 1.25% only (see Fig. 3 and Table 2).

For $\lambda>>1$ we obtain instead of (5.13):

$$E_1^{(0)}=\frac{9}{8}(10)^{\frac{1}{3}}\lambda^{\frac{1}{3}}+\frac{3}{4}(10)^{-\frac{1}{3}}\lambda^{-\frac{1}{3}}+\cdots\cdots$$

$$E_1^{(2)}=\frac{9}{8}(10)^{\frac{1}{3}}\left(-\frac{1}{93}\right)\lambda^{\frac{1}{3}}+\frac{3}{4}(10)^{-\frac{1}{3}}\left(\frac{61}{2883}\right)\lambda^{-\frac{1}{3}}+\cdots\cdots \tag{5.18}$$

$$E_1^{(4)}=\frac{9}{8}(10)^{\frac{1}{3}}(-0.0017953)\lambda^{\frac{1}{3}}+\frac{3}{4}(10)^{-\frac{1}{3}}(0.0069213)\lambda^{-\frac{1}{3}}+\cdots\cdots$$

$$\bar{E}_1^{(4)}=E_1^{(0)}+E_1^{(2)}+E_1^{(4)}=2.3933259\,\lambda^{\frac{1}{3}}+0.3578943\,\lambda^{-\frac{1}{3}}+\cdots \tag{5.18a}$$

This should be compared to the exact values (cf. 1-Table 1)

$$E_1^{(e)}=2.39364402\,\lambda^{\frac{1}{3}}+0.35780\,\lambda^{-\frac{1}{3}}+\cdots\cdots$$

For small λ our α_1 is given by

$$\alpha_1=1-5\lambda+\frac{5}{2}(5\lambda)^2-8(5\lambda)^3+\frac{231}{8}(5\lambda)^4+\cdots\cdots \tag{5.19}$$

and we get

$$\left.\begin{aligned}
E_1^{(0)} &= \frac{3}{2} + \frac{3}{4}(5\lambda) - \frac{3}{4}(5\lambda)^2 + \frac{3}{2}(5\lambda)^3 - \frac{63}{16}(5\lambda)^4 + \cdots \\
E_1^{(2)} &= \qquad\qquad - \frac{3}{40}(5\lambda)^2 + \frac{183}{400}(5\lambda)^3 - \frac{2193}{4000}(5\lambda)^4 + \cdots \\
E_1^{(4)} &= \qquad\qquad\qquad\qquad\qquad\qquad - \frac{33717}{256000}(5\lambda)^4 + \cdots
\end{aligned}\right\} \quad (5.20)$$

$\therefore$

$$E_1^{(0)} = \frac{3}{2} + \frac{15}{4}\lambda + \cdots$$

$$E_1^{(0)} + E_1^{(2)} = \frac{3}{2} + \frac{15}{4}\lambda - \frac{165}{8}\lambda^2 + \frac{3915}{16}\lambda^3 + \cdots \qquad (5.21)$$

$$E_1^{(0)} + E_1^{(2)} + E_1^{(4)} = \frac{3}{2} + \frac{15}{4}\lambda - \frac{165}{8}\lambda^2 + \frac{3915}{16}\lambda^3 - \frac{5910345}{2048}\lambda^4 + \cdots$$

We believe these to be also the correct coefficients and the discussion given for i=0 is also valid here, since $E_1^{(3)} = 0$ and $E_1^{(5)} \neq 0$.

§6. The Small λ Regime for Higher Excited States ($i \geq 2$)

As the calculation of perturbations due to $H_i^{(I)}$ higher than the 3rd order becomes quite complicated for $i \geq 2$, we will be content with our 2nd order perturbation formulae (3.1), (3.2) and (4.4). Since these formulae are the basis for the following three sections, we recapitulate them here

$$E_i^{(0)} = \left(i + \frac{1}{2}\right) \frac{1}{4}\left(\frac{3}{\alpha_i} + \alpha_i\right)$$

$$E_i^{(2)} = \left(\frac{1-\alpha_i^2}{6}\right)^2 \frac{1}{\alpha_i(2i^2+2i+1)}$$

$$\times \left[\left(\frac{2i+1}{4}\right)^2 \left\{ -\frac{(i+4)(i+3)(i+2)(i+1)}{8i^2+16i+7-\alpha_i^2(8i+3)} + \frac{i(i-1)(i-2)(i-3)}{8i^2-1+\alpha_i^2(8i+5)} \right\} \right. \tag{6.1}$$

$$\left. +2(i+2)(i+1)i(i-1)\left\{ -\frac{i(i-1)}{8i^2+12i+5-\alpha_i^2(4i+1)} + \frac{(i+2)(i+1)}{8i^2+4i+1+\alpha_i^2(4i+3)} \right\} \right]$$

where α_i is the positive real root of

$$y_i \alpha_i^3 + \alpha_i^2 - 1 = 0 \quad , \quad y_i = 6\lambda\left(\frac{2i^2+2i+1}{2i+1}\right) \tag{6.2}$$

For small λ the solution of (6.2) is given by

$$\alpha_i = 1 - \frac{y_i}{2} + \frac{5}{2}\left(\frac{y_i}{2}\right)^2 + \cdots\cdots \tag{6.3}$$

We see that $(1-\alpha_i^2)$ is proportional to y_i, or λ, so the factor $(1-\alpha_i^2)^2$ in front of $E_i^{(2)}$ is proportional to λ^2.

Hence for small λ we retain only terms proportional to λ in the rest of $E_i^{(2)}$. We thus can rewrite $E_i^{(2)}$ in the following form

$$E_i^{(2)} \approx -\lambda^2 \left(\frac{1}{\alpha_i^2}\right)\left[\frac{(i+4)(i+3)(i+2)(i+1)}{16\left[4+(2i+5)6\lambda\right]} - \frac{i(i-1)(i-2)(i-3)}{16\left[4+(2i-3)6\lambda\right]}\right. \tag{6.4}$$

$$\left. + \frac{(i+2)(i+1)\,i(i-1)}{(2i+1)^2}\left\{\frac{i(i-1)}{2+(2i+3)3\lambda} - \frac{(i+2)(i+1)}{2+(2i-1)3\lambda}\right\}\right]$$

$$(\lambda \ll 1)$$

Together with $E_i^{(0)}$ for small λ

$$E_i^{(0)} \approx \left(i+\frac{1}{2}\right) + \frac{3}{4}\left[1+2i(i+1)\right]\lambda - \frac{9}{4}\frac{(2i^2+2i+1)^2}{2i+1}\lambda^2 + \cdots \tag{6.5}$$

we obtain a formula for $(E_i^{(0)}+E_i^{(2)})$, which resembles the formula (1-1.17) if we put $\frac{1}{\alpha_i^2}$ equal to 1. But $\frac{1}{\alpha_i^2}$ is equal to 1 plus terms proportional to λ, so that it is a little bit arbitrary to keep λ otherwise and not in α_i. Still (6.4) is not a consistent expansion in powers of λ. If we neglect consistently all terms proportional to λ^3, then we get

$$E_i^{(2)} \approx \frac{4i^4+8i^3-25i^2-29i-3}{8(2i+1)}\lambda^2 + \cdots$$

and finally we obtain

$$E_i^{(0)} + E_i^{(2)} \approx \left(i+\frac{1}{2}\right) + \frac{3}{4}(2i^2+2i+1)\lambda \tag{6.6}$$

$$- \frac{34i^3+51i^2+59i+21}{8}\lambda^2 + \cdots$$

This agrees with the results for i=0,1 in §5 up to order λ^2, as it should be.

Now if we let $i\to\infty$, but keep $\lambda i=\beta$ finite and small, we get

$$\frac{E_i^{(0)}+E_i^{(2)}}{i+\frac{1}{2}} \approx 1+\frac{3}{2}\beta-\frac{17}{4}\beta^2+\cdots\cdots \tag{6.7}$$

This agrees with the formula (1-1.18b) for small β. We retained terms up to λ^2, so that in the coefficient of β^2 β should be put to zero. We will give the formulae valid for any β in §8. The formula (1-1.18b) is not valid for $\beta\gtrsim 1$, as will be seen in §8, while our formula (8.3) is valid also for large β.

§7. The Large λ Regime for Higher Excited States (i≥2)

For i=0,1, we already gave a fourth order calculation in §5. We use here also our second order formulae summarized in (6.1), (6.2), and expand them for large λ. In this case α_i is given by

$$\left.\begin{aligned} \alpha_i &\approx (y_i)^{-\frac{1}{3}}\left(1-\frac{1}{3}y_i^{-\frac{2}{3}}+\frac{1}{9}y_i^{-\frac{4}{3}}+\cdots\right) \\ y_i &= 6\lambda\left(\frac{2i^2+2i+1}{2i+1}\right) \end{aligned}\right\} \qquad (7.1)$$

So, if we expand in powers of $y_i^{-\frac{1}{3}}$ (or $\lambda^{-\frac{1}{3}}$), we get

$$E_i^{(0)} \approx \left(i+\frac{1}{2}\right)\frac{1}{4}(y_i)^{\frac{1}{3}}\left(3+2y_i^{-\frac{2}{3}}-\frac{1}{3}y_i^{-\frac{4}{3}}+\cdots\right)$$

$$\begin{aligned} E_i^{(2)} \approx\ & \frac{\left(1-y_i^{-\frac{2}{3}}+\frac{2}{3}y_i^{-\frac{4}{3}}\right)^2}{36\left(1-\frac{1}{3}y_i^{-\frac{2}{3}}+\frac{1}{9}y_i^{-\frac{4}{3}}\right)}\cdot\frac{y_i^{\frac{1}{3}}}{(2i^2+2i+1)} \\ & \times\left[\left(\frac{2i+1}{4}\right)^2\left\{-\frac{(i+4)(i+3)(i+2)(i+1)}{8i^2+16i+7-(8i+3)y_i^{-\frac{2}{3}}\left(1-\frac{2}{3}y_i^{-\frac{2}{3}}\right)}\right.\right. \qquad (7.2) \\ & \qquad\left.+\frac{i(i-1)(i-2)(i-3)}{8i^2-1+(8i+1)y_i^{-\frac{2}{3}}\left(1-\frac{2}{3}y_i^{-\frac{2}{3}}\right)}\right\} \\ & +2(i+2)(i+1)i(i-1)\left\{-\frac{i(i-1)}{8i^2+12i+5-(4i+1)y_i^{-\frac{2}{3}}\left(1-\frac{2}{3}y_i^{-\frac{2}{3}}\right)}\right. \\ & \qquad\left.\left.+\frac{(i+2)(i+1)}{8i^2+4i+1+(4i+3)y_i^{-\frac{2}{3}}\left(1-\frac{2}{3}y_i^{-\frac{2}{3}}\right)}\right\}\right] \end{aligned}$$

If we expand our result in the form

$$E_i^{(0)} + E_i^{(2)} \approx \lambda^{\frac{1}{3}} \left(\varepsilon_i^{Y} + \alpha_i^{Y} \lambda^{-\frac{2}{3}} + \beta_i^{Y} \lambda^{-\frac{4}{3}} + \cdots \right) \tag{7.3}$$

as in reference 1, our formula for β_i^Y becomes too elaborate, so we will only give our expression for ε_i^Y and α_i^Y ;

$$\begin{aligned} \varepsilon_i^{Y} = \left(\frac{6(2i^2+2i+1)}{2i+1}\right)^{\frac{1}{3}} \Bigg[& \frac{3}{4}\left(i+\frac{1}{2}\right) \\ & + \frac{1}{36(2i^2+2i+1)} \Bigg\{ \left(\frac{2i+1}{4}\right)^2 \left(-\frac{(i+4)(i+3)(i+2)(i+1)}{8i^2+16i+7} + \frac{i(i-1)(i-2)(i-3)}{8i^2-1} \right) \\ & + 2(i+2)(i+1)i(i-1)\left(-\frac{i(i-1)}{8i^2+12i+5} + \frac{(i+2)(i+1)}{8i^2+4i+1} \right) \Bigg\} \Bigg] \end{aligned} \tag{7.4a}$$

$$\begin{aligned} \alpha_i^{Y} = \left(\frac{6(2i^2+2i+1)}{2i+1}\right)^{-\frac{1}{3}} \Bigg[& \frac{1}{2}\left(i+\frac{1}{2}\right) \\ & - \frac{1}{36(2i^2+2i+1)} \Bigg\{ \left(\frac{2i+1}{4}\right)^2 \Bigg(\frac{(i+4)(i+3)(i+2)(i+1)}{8i^2+16i+7} \left(\frac{8i+3}{8i^2+16i+7} - \frac{5}{3} \right) \\ & + \frac{i(i-1)(i-2)(i-3)}{8i^2-1} \left(\frac{8i+5}{8i^2-1} + \frac{5}{3} \right) \Bigg) \\ & + 2(i+2)(i+1)i(i-1) \Bigg(\frac{i(i-1)}{8i^2+12i+5} \left(\frac{4i+1}{8i^2+12i+5} - \frac{5}{3} \right) \\ & + \frac{(i+2)(i+1)}{8i^2+4i+1} \left(\frac{4i+3}{8i^2+4i+1} + \frac{5}{3} \right) \Bigg) \Bigg\} \Bigg] \end{aligned} \tag{7.4b}$$

This formula can be compared with the corresponding numerical values given in Ref. 1, Table 1 and 2, which will be done in §9 and is summarized in our Table 4. Contrary to the case of small λ in the previous section, where "our" perturbation theory contains the "usual" perturbation theory, here our higher order contributions $E_i^{(3)}$, $E_i^{(4)}$ ..... always bring a small numerical correction also for the coefficients of the leading terms (proportional to $\lambda^{\frac{1}{3}}$, $\lambda^{-\frac{1}{3}}$....).

§8. The Large i Regime for Arbitrary Finite iλ

For large i, it is more convenient to use

$$j = i + \frac{1}{2} \tag{8.1}$$

as variable. Then our second order perturbation results (6.1) have a more symmetrical form:

$$\varepsilon_i^{(0)} \equiv \frac{E_i^{(0)}}{j} = \frac{1}{4}\left(\frac{3}{\alpha_i} + \alpha_i\right)$$

$$\frac{E_i^{(2)}}{j} = \left(\frac{1-\alpha_i^2}{6}\right)^2 \frac{1}{2\alpha_i(j^2+\frac{1}{4})}$$

$$\times\left[\left(\frac{j}{2}\right)^2\left\{-\frac{(j^2+4j+\frac{7}{4})(j^2+4j+\frac{15}{4})}{8j^2+8j+1-\alpha_i^2(8j-1)} + \frac{(j^2-4j+\frac{7}{4})(j^2-4j+\frac{15}{4})}{8j^2-8j+1+\alpha_i^2(8j+1)}\right\}\right. \tag{8.2}$$

$$\left. +2(j^2-\tfrac{1}{4})(j^2-\tfrac{9}{4})\left\{-\frac{(j^2-2j+\frac{3}{4})}{8j^2+4j+1-\alpha_i^2(4j-1)} + \frac{(j^2+2j+\frac{3}{4})}{8j^2-4j+1+\alpha_i^2(4j+1)}\right\}\right]$$

where

$$y_i\alpha_i^3 + \alpha_i^2 - 1 = 0\,, \quad y_i = 6\lambda\left(\frac{2i^2+2i+1}{2i+1}\right) = 6\lambda j\left(1+\frac{1}{4j^2}\right)$$

After some algebra we arrive at

$$\frac{E_i^{(2)}}{j} = \left(\frac{1-\alpha_i^2}{12}\right)^2 \frac{1}{\alpha_i}\left(\frac{4j^2}{4j^2+1}\right)\left[\frac{32(j^2-\frac{1}{16j^2})(j^2-\frac{9}{4})(5-\alpha_i^2)}{(8j^2+1+\alpha_i^2)^2 - 16j^2(1-\alpha_i^2)^2}\right.$$

$$\left. - \frac{8j^4(7+\alpha_i^2)+4j^2(1+45\alpha_i^2)-\frac{1}{2}(61-149\alpha_i^2)}{(8j^2+1+\alpha_i^2)^2 - 64j^2(1-\alpha_i^2)^2}\right]$$

This is still valid for any $j(= i + \frac{1}{2})$.
Now let $j\to\infty$, keeping $j\lambda\equiv\Lambda$ finite, large or small. For any i and λ, α_i is always between 0 and 1, as can be clearly seen from Fig. 3. Then we get

$$\frac{E_i^{(2)}}{j} \approx \frac{1}{8\alpha_i}\left(\frac{1-\alpha_i^2}{12}\right)^2\left[(13-5\alpha_i^2) - \frac{1}{4j^2}(216+85\alpha_i^2-17\alpha_i^4+8\alpha_i^6)+\cdots\right]^{+)}$$

So our formula for large j is given by

$$\varepsilon_i^{(2)} = \frac{E_i^{(0)}+E_i^{(2)}}{(i+\frac{1}{2})} = \frac{1}{4}\left(\frac{3}{\alpha_i}+\alpha_i\right) + \frac{1}{8\alpha_i}\left(\frac{1-\alpha_i^2}{12}\right)^2(13-5\alpha_i^2) \qquad (8.3)$$

This is our formula for large i and any finite $\Lambda = (i + \frac{1}{2})\lambda$. α_i is given as usual, which covers quite well large and small Λ. For small $\Lambda(\sim y_i)$ we use (6.3), and inserting this into (8.3) we get (6.5), as it should be. For large $\Lambda(\sim y_i)$ we use (7.1), inserting this into (8.3) and expanding

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+) Now we neglect the $(\frac{1}{4j^2})$-term. In view of the large numerical coefficient one may be afraid that this neglection could be dangerous for some range of finite $\Lambda=j\lambda$. As we have assumed j large and Λ finite, small Λ ($\alpha_i\to 1$) means j exceedingly large, so that the $1/4j^2$ -term can be safely neglected in spite of its large numerical coefficient. And for large $\Lambda(\alpha_i\to 0)$ we neglected $216/4j^2$ in comparison to 13, which is also all right for, say, $j>10$ because $E_i^{(2)}$ itself is only a small correction to $E_i^{(0)}$.

up to the power $\Lambda^{-\frac{2}{3}}$ we get

$$E_i^{(0)} + E_i^{(2)} \approx \lambda^{\frac{1}{3}} \Big[\frac{(6)^{\frac{1}{3}}}{8} \Big(\frac{877}{144}\Big)(i+\tfrac{1}{2}) \Big\{ i + \tfrac{1}{2} + \frac{1}{4(i+\frac{1}{2})} \Big\}^{\frac{1}{3}}$$

$$+ \frac{1}{8(6)^{\frac{1}{3}}} \Big(\frac{103}{27}\Big)(i+\tfrac{1}{2}) \Big\{ i + \tfrac{1}{2} + \frac{1}{4(i+\frac{1}{2})} \Big\}^{-\frac{1}{3}} \lambda^{-\frac{2}{3}} + \cdots\cdots \Big]$$

$$= \lambda^{\frac{1}{3}} \Big[1.383346 (i+\tfrac{1}{2}) \Big(i + \tfrac{1}{2} + \frac{1}{4(i+\frac{1}{2})} \Big)^{\frac{1}{3}}$$

$$+ 0.262422 (i+\tfrac{1}{2})^{\frac{2}{3}} \Big(1 + \frac{1}{4(i+\frac{1}{2})^2} \Big)^{-\frac{1}{3}} \lambda^{-\frac{2}{3}} + \cdots\cdots \Big]$$

$$\approx \lambda^{\frac{1}{3}} \Big[1.383346 \Big(i + \tfrac{1}{2} + \frac{1}{16(i+\frac{1}{2})} \Big)^{\frac{4}{3}}$$

$$+ 0.262422 (i+\tfrac{1}{2})^{\frac{2}{3}} \lambda^{-\frac{2}{3}} + \cdots\cdots \Big] \qquad (8.4)$$

This should be compared with the more accurate value (2-IV.16a)

$$E_i^{(e)} = \lambda^{\frac{1}{3}} \Big[1.376507 \Big(i + \tfrac{1}{2} + \frac{0.0265}{i+\frac{1}{2}} \Big)^{\frac{4}{3}}$$

$$+ 0.268055 (i+\tfrac{1}{2})^{\frac{2}{3}} \lambda^{-\frac{2}{3}} + \cdots\cdots \Big]$$

Except that we get $\frac{1}{16}$ = 0.0625 in place of 0.0265 in the first term, the agreement here is also satisfactory, especially in view of the fact, that we used only "our" second order perturbation formula.

Thus our formula (8.3) for large i covers really well all ranges of finite $\Lambda = (i+\frac{1}{2})\lambda$. The formula (1-1.18b) covers only small Λ, whereas our formula (8.3) [which gives (6.5) in this case], is equally well suited and, as we have shown in Fig. 3, our formula (8.3) and (1-1.18b) deviate near $\Lambda \sim 1$, as expected. Our formula covers the intermediate Λ region also well and connects smoothly with the correct result for large Λ.

§9. Intermediate Regime of i and λ. Numerical Results.

For intermediate magnitudes of i and λ we must return to formula (3.1) for our lowest order, and (4.4) for our second order corrected results. First, for i=0, we obtain from our forth order formula (5.12) the numerical results summarized in Table 1. For this calculation we still need only a desk computer. For i=1, from the corresponding formula (5.17), we obtain the results of Table 2. As will be seen the maximal errors (in the case of large λ) are less than 0.012% and 0.013%, respectively.

For the states with $i \neq 0,1$ we must be content with the much more inaccurate formula (4.4). We have listed as examples the lowest [(3.1)] and second order [(4.4)] results for i=2,3 and 8 in Table 3. The accuracy of our approximation procedure can be most clearly seen from Fig. 4, where we plot the ratio of our results to the exact values as a function of λ. We see that in general

$$E_i^{(0)} < E_i^{(e)} < \bar{E}_i^{(2)} = E_i^{(0)} + E_i^{(2)}, \text{ and } |\bar{R}_i^{(2)} - 1| = \left|\frac{\bar{E}_i^{(2)} - E_i^{(e)}}{E_i^{(e)}}\right|$$

are much smaller than $|\bar{R}_i^{(0)} - 1| = \left|\frac{\bar{E}_i^{(0)} - E_i^{(e)}}{E_i^{(e)}}\right|$ for any i.

This means "our" 2nd order perturbation calculations improve "our" lowest order results essentially, for all ranges of λ. In Table 4 we have summarized our calculated $E_i^{(0)}$ and $\bar{E}_i^{(2)}$ for the limiting case of large λ [eqs. (7.3),(7.4a),(7.4b)], and even for this case we see that the agreement is satisfactory.

§10. Concluding Remarks and Discussion.

As we have seen, our method and the required numerical calculations are extremely simple, they contain only one adjustable parameter α_i for given i and λ in the special combination $y_i = 6\lambda(2i^2+2i+1)/$ $/(2i+1)$. We see that the largest error occurs at $\lambda\to\infty$ and i=0. The other extreme case, $i\to\infty$, is less inaccurate. And even for the ($\lambda\to\infty$, i=0)-case "our" fourth order perturbation reduces the error to 0.01%. From these numerical results we are quite certain that if we calculate $E_i^{(3)}$ and $E_i^{(4)}$ also for $i\geq 2$, then everything is expected to agree within, say, about 0.01%.

As our method is simple enough, we may apply our method also to a pair of coupled oscillators with quartic coupling, as discussed in reference 1. We have to introduce two parameters α_1 and α_2 for each oscillator. Our preliminary calculations show that our approach is also successful in this case. We will discuss these problems in a forthcoming paper.

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Figure captions

Fig. 1. Our lowest order energy value

$\varepsilon_i^{(0)} = E_i^{(0)}/(i+\frac{1}{2})$ of equ. (3.1) as a function

of $y_i = 6\lambda \frac{(2i^2+2i+1)}{2_i}$ on a logarithmic scale.

Fig. 2. Our $\varepsilon_i^{(0)}$ of equ. (3.1) for small y and some exact values, $E_i(0.1)$ means $E_i^{(e)}$ for λ=0.1, for example.

Fig. 3. Our characteristic variational parameter α_i as a function of $y_i = 6\lambda \frac{(2i^2+2i+1)}{2i+1}$ on a logarithmic scale.

Fig. 4. Ratio $R_i^{(j)} = E_i^{(j)}/E_i^{(e)}$ as function of λ, where (j) represents the order of our approximation, and $E_i^{(e)}$ are the exact values, taken from reference 2.

Fig. 5. Comparison of various $\varepsilon_i = E_i/(i+\frac{1}{2})$ for very large i, finite $\beta=i\lambda$.
Equ. (6.5) is good only for small β, (1-1.18b) is better, but it becomes proportional to $y(\equiv 6\beta)$ for large β. Our formulae for $\varepsilon_i^{(0)}$ [(8.2)] and $\varepsilon_i^{(2)}$ [(8.3)] deviate at about β = 1(y=6) from (1-1.18b) and are valid for intermediate <u>and</u> large β.

Table 1. The Ground State Energies $E_o^{(j)}$, eq. (5.12)

λ	α_o	$E_o^{(0)}$	$-E_o^{(2)}$	$-E_o^{(4)}$	$\bar{E}_o^{(2)}$	$\bar{E}_o^{(4)}$	$E_o^{(e)}$
0.002	0.994088	0.501491	0.000001	0.000000	0.501490	0.501490	0.501490
0.010	0.972055	0.507288	31	0	0.507257	0.507257	0.507256
0.10	0.818869	0.560308	0.001107	58	0.559201	0.559143	0.559146
0.5	0.598194	0.701661	4847	365	0.638337	0.637972	0.637992
1.0	0.500000	0.812500	7500	1255	0.805000	0.803745	0.803771
2.0	0.410704	0.964405	0.010797	0.002050	0.953609	0.951560	0.951568
50	0.148277	2.547585	38763	8898	2.508822	2.499924	2.499709
200	0.093827	4.008446	62564	0.014580	3.945882	2.931302	3.930331
1000	0.054977	6.82791	0.107757	25277	6.720154	6.694877	6.694221
8000	0.027509	13.63534	216122	50826	13.41922	13.36839	13.36691
20000	0.020271	18.50187	0.293451	0.069039	18.20842	18.13938	18.13723

Table 2. The First Excited State Energies $E_1^{(j)}$. eq. (5.17)

λ	α_1	$E_1^{(0)}$	$-E_1^{(2)}$	$-E_1^{(4)}$	$\bar{E}_1^{(2)}$	$\bar{E}_1^{(4)}$	$E_1^{(e)}$
0.002	0.990242	1.507427	0.000007	0.000000	1.507420	1.507420	1.507419
0.01	0.955401	1.535792	142	0	1.535649	1.535649	1.535648
0.10	0.754878	1.773387	3717	186	1.769670	1.769484	1.769503
0.5	0.525166	2.339117	0.013391	1457	2.325726	2.324269	2.324406
1.0	0.433105	2.759937	19749	2516	2.740188	2.737672	2.737892
2.0	0.352469	3.323946	27536	3876	3.296411	3.292535	3.292868
50	0.125329	9.023373	94359	0.015390	8.929014	8.913624	8.915096
200	0.079204	14.23353	0.151155	25039	14.08238	14.05734	14.05923
1000	0.046383	24.27197	0.259880	43274	24.01209	24.96881	23.97221
8000	0.023204	48.49172	520862	86907	47.97085	47.88395	47.89077
20000	0.017098	65.80358	0.707157	0.118028	65.09642	64.97839	64.98668

Comparison of our calculated values with the exact values $E_i^{(e)}$, taken from Table IIA of reference 2. $E_i^{(j)}$ is our j-th order calculation, and $\bar{E}_i^{(2)} = E_i^{(0)}+E_i^{(2)}$, $\bar{E}_i^{(4)} = E_i^{(0)}+E_i^{(2)}+E_i^{(4)}$ $(i = 0,1)$.

Table 3. E_2, E_3, E_8, eq. (4.4).

λ	$\varepsilon_2^{(0)}$	$\bar{\varepsilon}_2^{(2)}$	$\varepsilon_2^{(e)}$	$\varepsilon_3^{(0)}$	$\bar{\varepsilon}_3^{(2)}$	$\varepsilon_3^{(e)}$	$\varepsilon_8^{(0)}$	$\bar{\varepsilon}_8^{(2)}$	$\varepsilon_8^{(e)}$
0.002	1.007682	1.007681	1.007681	1.010497	1.010501	1.010498	1.024397	1.024451	1.024449
0.006	1.022396	1.022390	1.022389	1.030305	1.030343	1.030339	1.067594	1.067961	1.067924
0.01	1.036354	1.036343	1.036338	1.048799	1.049775	1.048884	1.105504	1.106310	1.106199
0.05	1.149601	1.149729	1.149592	1.192308	1.193240	1.193450	1.362349	1.367747	1.366444
0.1	1.255298	1.255850	1.255450	1.320552	1.323360	1.322538	1.567474	1.577016	1.573996
0.3	1.536962	1.539270	1.537913	1.651281	1.658635	1.656164	2.059991	2.079055	2.073229
0.5	1.729408	1.733072	1.731010	1.872784	1.883163	1.879543	2.376580	2.401351	2.393552
0.7	1.882003	1.886750	1.885312	2.047043	2.059733	2.055219	2.621785	2.650804	2.641529
1	2.068951	2.075005	2.071717	2.259407	2.274828	2.269258	2.917599	2.951587	2.940582
2	2.517306	2.526375	2.521552	2.765877	2.787501	2.779235	3.614370	3.659794	3.644692
50	6.958084	6.992653	6.974796	7.723295	7.797861	7.769327	10.29475	10.43847	10.38992
200	10.99965	11.05571	11.02057	12.21466	12.33464	12.28722	16.30700	16.53669	16.45887
1000	18.76000	18.85676	18.80694	20.84986	21.05620	20.97689	27.85878	28.25269	28.11901
8000	37.49001	37.68422	37.58424	41.67348	42.08710	41.92729	55.69919	56.48792	56.21832
20000	50.87529	51.13902	51.00360	56.56014	57.12177	56.89861	75.59208	76.66275	76.29592

$$E_i^{(0)} = (i+\tfrac{1}{2})\varepsilon_i^{(0)}, \quad \bar{E}_i^{(2)} = (i+\tfrac{1}{2})(\varepsilon_i^{(0)}+\varepsilon_i^{(2)}) = (i+\tfrac{1}{2})\bar{\varepsilon}_i^{(2)}, \quad E_i^{(e)} = (i+\tfrac{1}{2})\varepsilon_i^{(e)}$$

$E_i^{(0)}$ [$\bar{E}_i^{(2)}$] are our calculated lowest [with second order correction] energy levels, from (3.1), (3.2) [(4.4)] . $E_i^{(e)}$ are exact values taken from Tables IIA, IIB, IIC of reference 2.

Table 4. $E_i^{(0)}$, $\bar{E}_i^{(2)}$ for $\lambda\to\infty$, (7.4a), (7.4b)

i	$\varepsilon_i^{y(0)}$	$\varepsilon_i^{y(2)}$	$\varepsilon_i^{(e)}$	$\alpha_i^{y(0)}/(i+\frac{1}{2})^{+\frac{2}{3}}$	$\alpha_i^{y(2)}/(i+\frac{1}{2})^{\frac{2}{3}}$	$\alpha_i^{(e)}/(i+\frac{1}{2})^{\frac{2}{3}}$
0	0.6814202	0.6706040	0.6679863	0.21840	0.22483	0.22806
1	2.423739	2.397677	2.393644	0.26566	0.27129	0.27305
2	4.684999	4.709304	4.696795	0.27159	0.26522	0.26816
3	7.291111	7.363548	7.335730	0.27331	0.26379	0.26820
4	10.166496	10.287822	10.244309	0.27404	0.26324	0.26813
5	13.267428	13.439319	13.379337	0.27441	0.26296	0.26810
6	16.564699	16.789097	16.711890	0.27462	0.26281	0.26808
7	20.037121	20.316011	20.220850	0.27475	0.26271	0.26808
8	23.668476	24.003803	23.889994	0.27484	0.26265	0.26807
9	27.445866	27.839509	27.706394	0.27491	0.26260	0.26807
10	31.358742	31.812501	31.659457	0.27495	0.26257	0.26806
100	1.36285	1.38335	1.376507	0.27516	0.26242	0.268055
101	$\times(i+\frac{1}{2})^{-4/3}$	$\times(i+\frac{1}{2})^{-4/3}$	$\times(i+\frac{1}{2})^{-4/3}$	0.27516	0.26242	0.268055

Captions of Table 4.

Eq. (7.4) valid for large λ.

$E_i^{(0)}/\omega = \lambda^{\frac{1}{3}}(\varepsilon_i^{y(0)}+\alpha_i^{y(0)}\lambda^{-\frac{2}{3}} + \ldots\ldots\ldots)$ ..... our lowest order result

$\bar{E}_i^{(2)}/\omega = \lambda^{\frac{1}{3}}(\varepsilon_i^{y(2)}+\alpha_i^{y(2)}\lambda^{-\frac{2}{3}} + \ldots\ldots\ldots)$ ..... with our second order corrections

$\varepsilon_i^{(e)}$ and $\alpha_i^{(e)}$ are the corresponding exact values, taken from Tables 1 and 2 of reference 1.

The author is indebted to C.B.Lang and L.Mathelitsch for computational aid in obtaining this table.

Fig. 1.

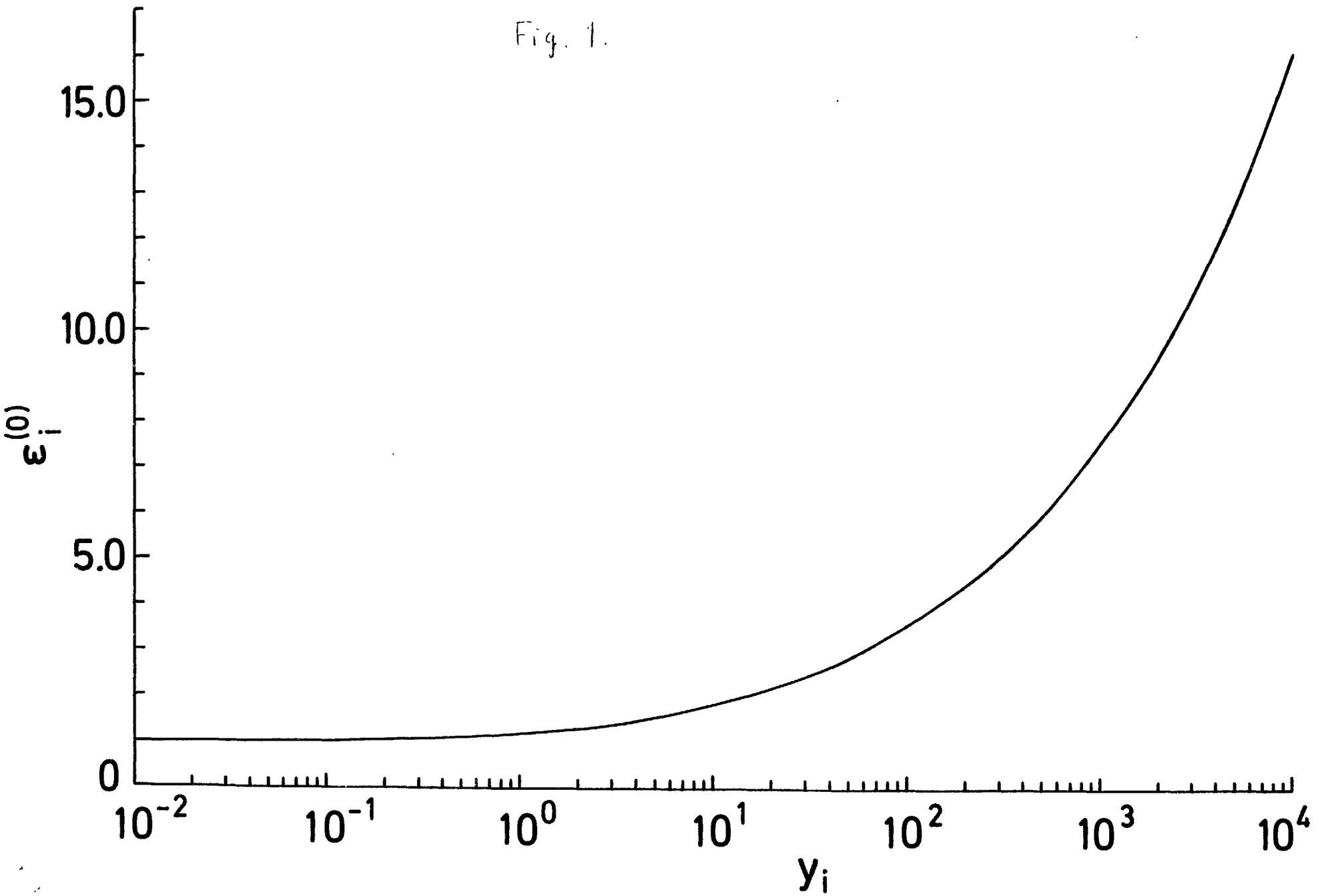


Fig 2

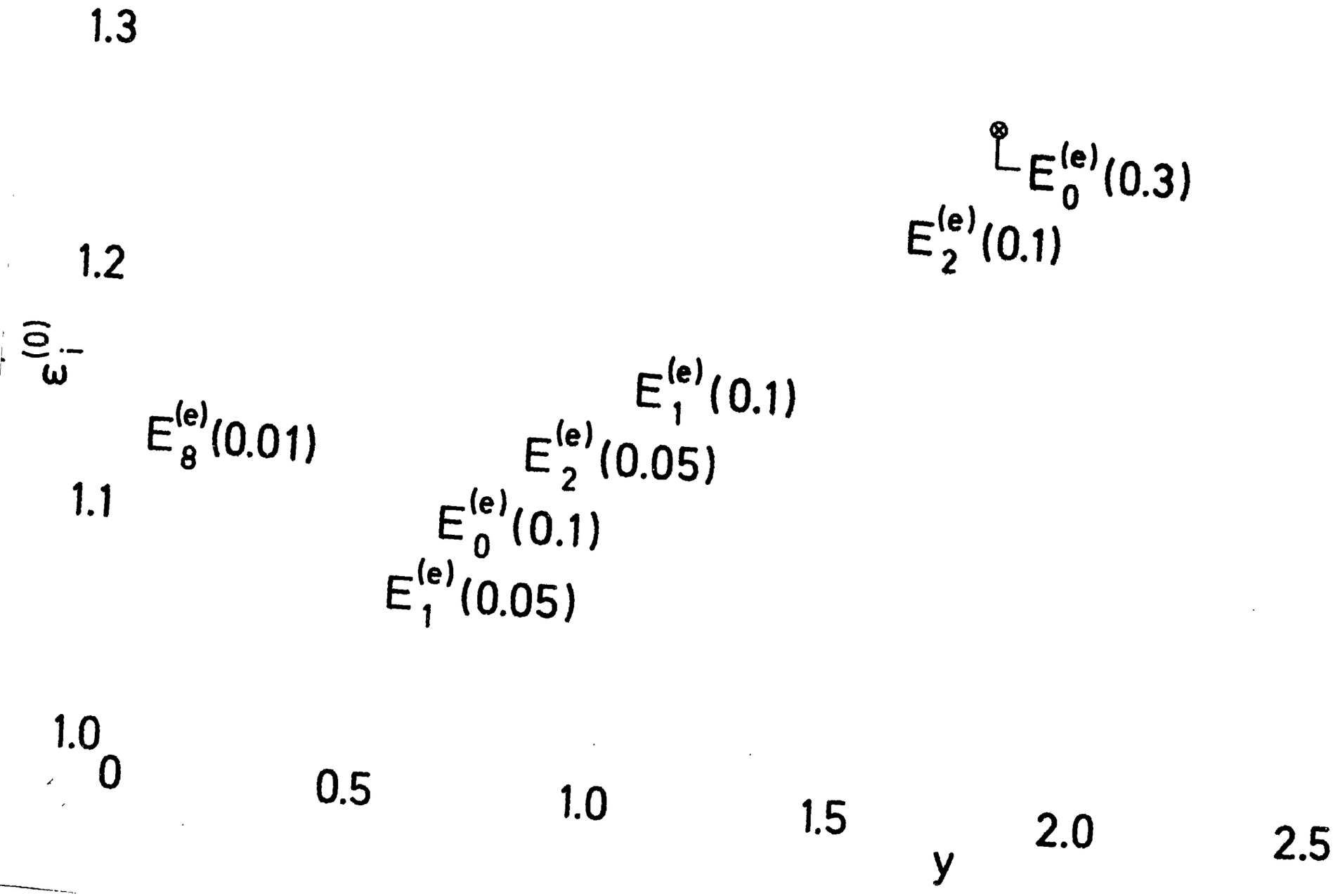

1.3
1.2
1.1
1.0
$\omega_i^{(0)}$
$E_0^{(e)}(0.3)$
$E_2^{(e)}(0.1)$
$E_1^{(e)}(0.1)$
$E_8^{(e)}(0.01)$
$E_2^{(e)}(0.05)$
$E_0^{(e)}(0.1)$
$E_1^{(e)}(0.05)$
0
0.5
1.0
1.5
2.0
2.5
y

Fig. 3

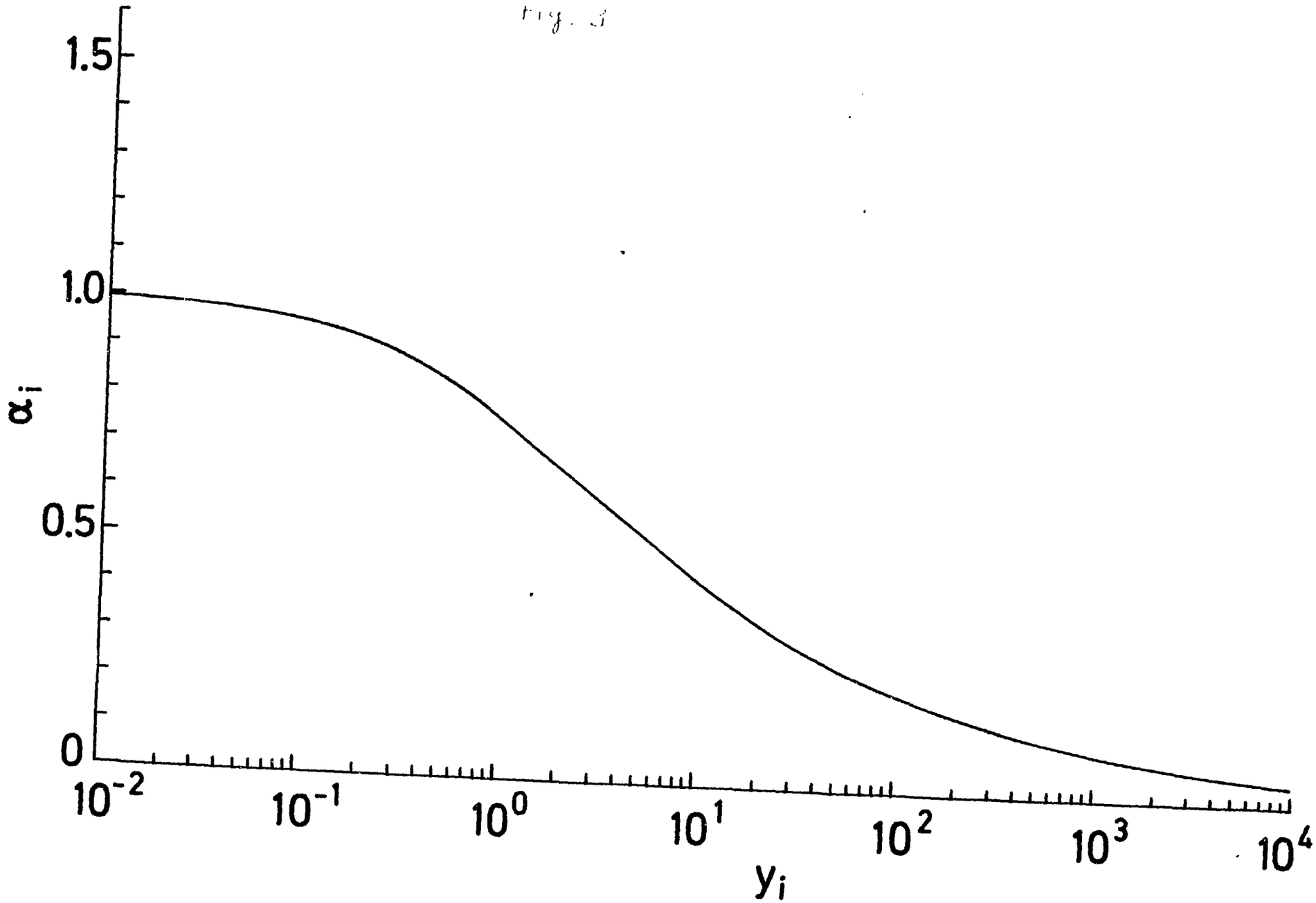
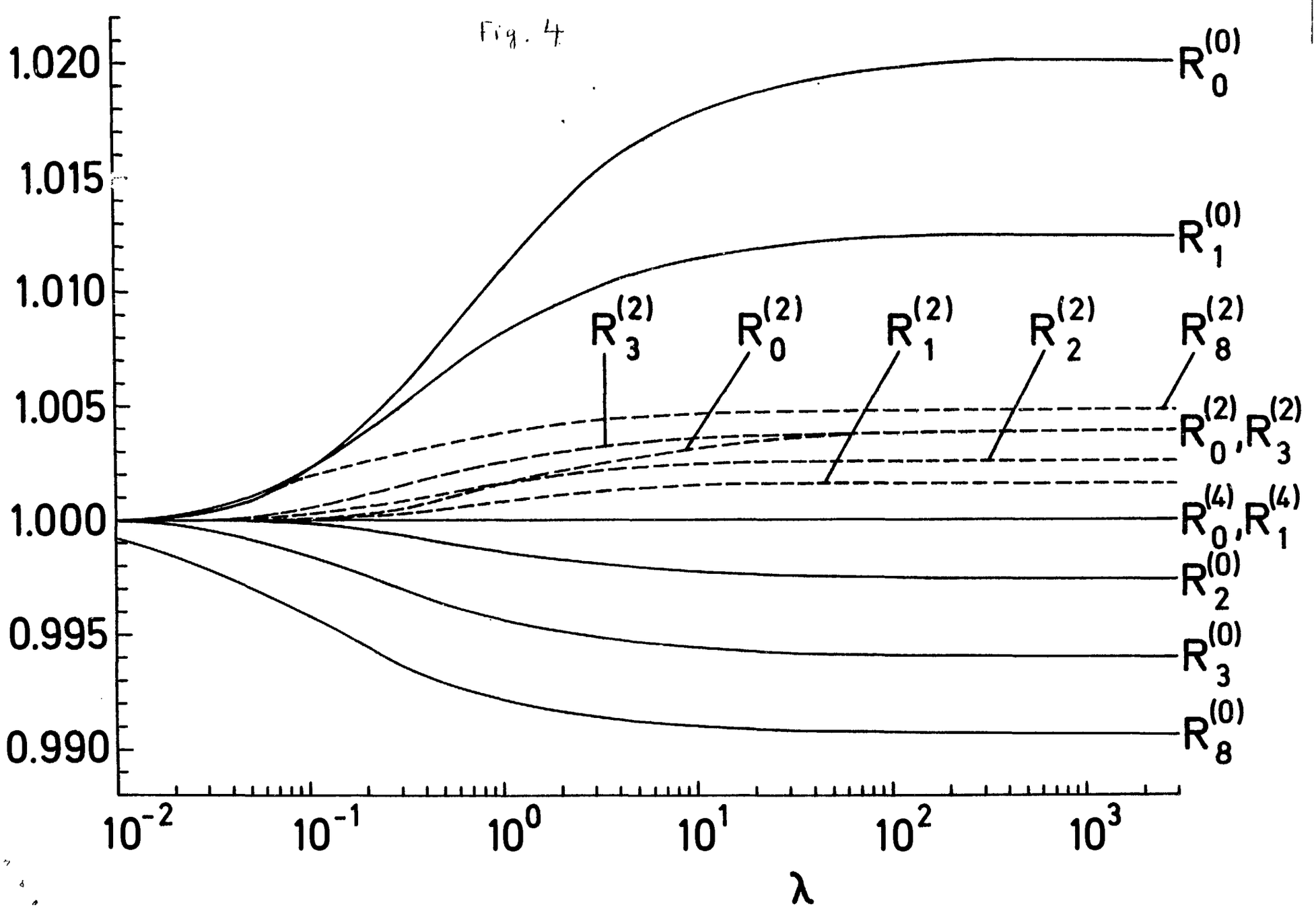


Fig. 4


1.020
1.015
1.010
1.005
1.000
0.995
0.990
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
λ
$R_0^{(0)}$
$R_1^{(0)}$
$R_3^{(2)}$
$R_0^{(2)}$
$R_1^{(2)}$
$R_2^{(2)}$
$R_8^{(2)}$
$R_0^{(2)}, R_3^{(2)}$
$R_0^{(4)}, R_1^{(4)}$
$R_2^{(0)}$
$R_3^{(0)}$
$R_8^{(0)}$

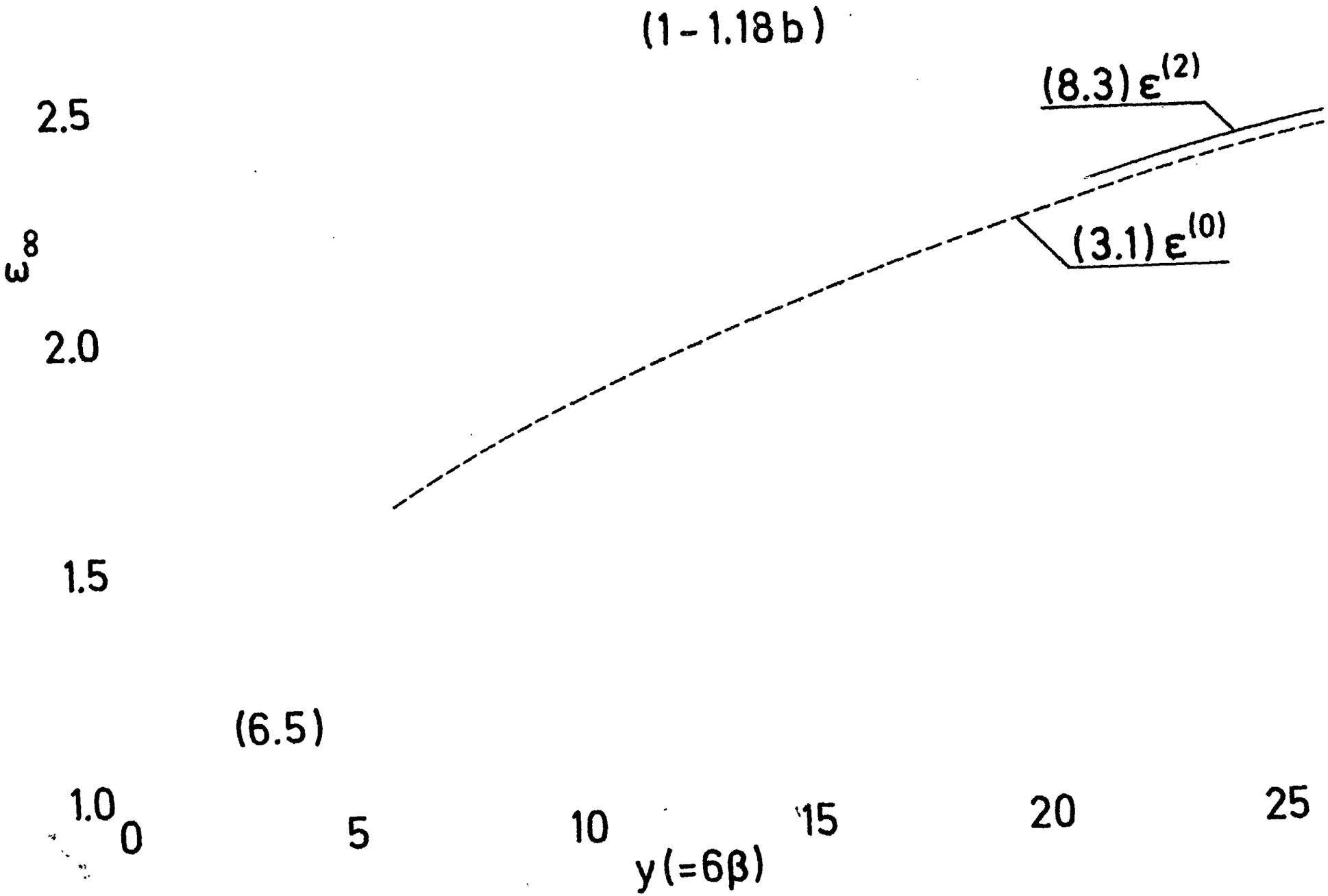



Fig. 5